1) a)

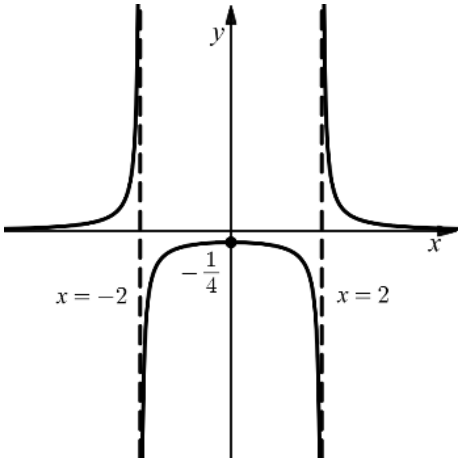
Asymptotes: $x = -2, x = 2$

Min at $(0, -4)$ becomes local max at $\left(0, -\frac{1}{4}\right)$

As $x \to -\infty$, $\frac{1}{\mathrm{f}(x)} \to 0^+$, as $x \to \infty$, $\frac{1}{\mathrm{f}(x)} \to 0^+$

As $x \to -2$ from LHS $\frac{1}{\mathrm{f}(x)} \to \infty$, as $x \to -2$ from RHS $\frac{1}{\mathrm{f}(x)} \to -\infty$

As $x \to 2$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to 2$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$



1) b)

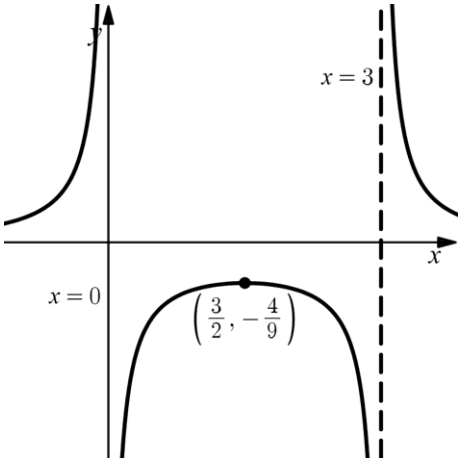
Asymptotes: $x = 0, x = 3$

Min at $\left(\frac{3}{2}, -\frac{9}{4}\right)$ becomes local max at $\left(\frac{3}{2}, -\frac{4}{9}\right)$

As $x \to -\infty$, $\frac{1}{\mathrm{f}(x)} \to 0^+$, as $x \to \infty$, $\frac{1}{\mathrm{f}(x)} \to 0^+$

As $x \to 0$ from LHS $\frac{1}{\mathrm{f}(x)} \to \infty$, as $x \to 0$ from RHS $\frac{1}{\mathrm{f}(x)} \to -\infty$

As $x \to 3$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to 3$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$



1) c)

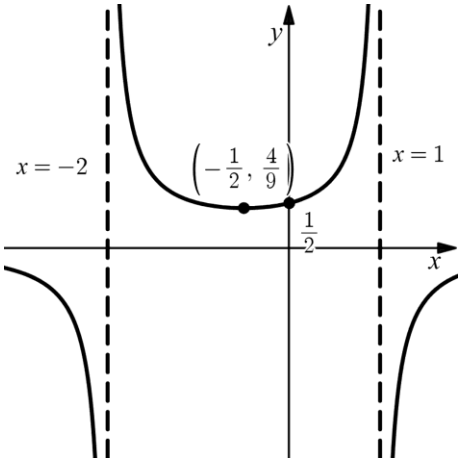
Asymptotes: $x = -2$, $x = 1$

Max at $\left(-\frac{1}{2}, \frac{9}{4}\right)$ becomes local min at $\left(-\frac{1}{2}, \frac{4}{9}\right)$

As $x \to -\infty$, $\frac{1}{\mathrm{f}(x)} \to 0^-$, as $x \to \infty$, $\frac{1}{\mathrm{f}(x)} \to 0^-$

As $x \to -2$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to -2$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$

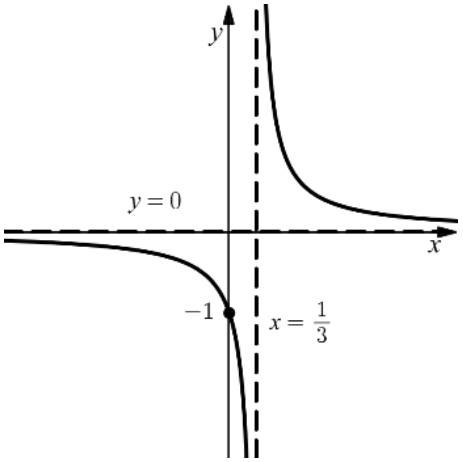
As $x \to 1$ from LHS $\frac{1}{\mathrm{f}(x)} \to \infty$, as $x \to 1$ from RHS $\frac{1}{\mathrm{f}(x)} \to -\infty$



2) a)

When $x = 0$, $y = -1$

Asymptotes: $x = \frac{1}{3}$, $y = 0$

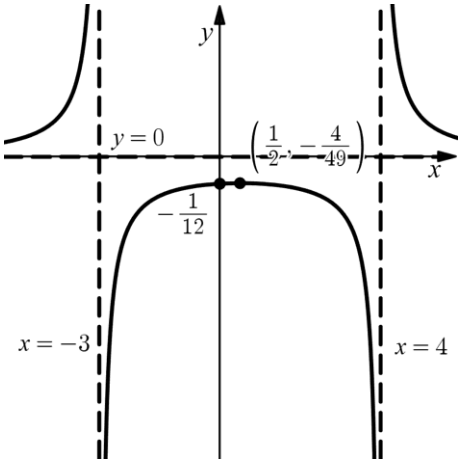


2) b)

When $x = 0$, $y = -\frac{1}{12}$

Asymptotes: $x = -3$, $x = 4$, $y = 0$

Min of $y = (x-4)(x+3)$ is at $x = \frac{1}{2}$, max of $y = \frac{1}{(x-4)(x+3)}$ is at $\left(\frac{1}{2}, -\frac{4}{49}\right)$

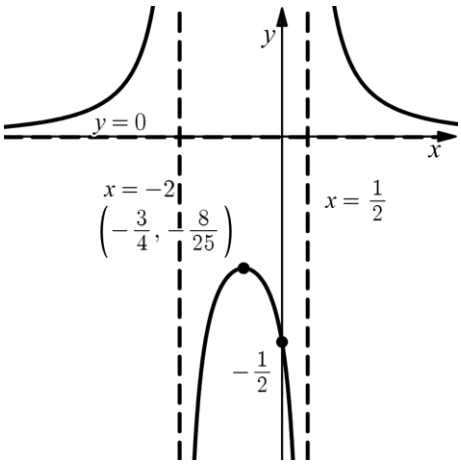


2) c)

When $x = 0$, $y = -\frac{1}{2}$

Asymptotes: $x = -2$, $x = \frac{1}{2}$, $y = 0$

Min of $y = 2x^2 + 3x - 2$ is at $x = -\frac{3}{4}$, max of $y = \frac{1}{2x^2+3x-2}$ is at $\left(-\frac{3}{4}, -\frac{8}{25}\right)$

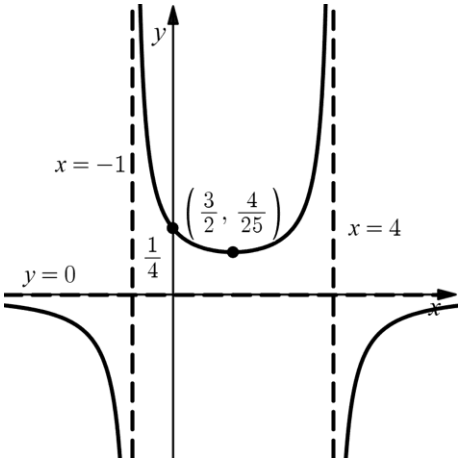


2) d)

When $x = 0$, $y = \frac{1}{4}$

Asymptotes: $x = -1$, $x = 4$, $y = 0$

Max of $y = 4 + 3x - x^2$ is at $x = \frac{3}{2}$, min of $y = \frac{1}{4+3x-x^2}$ is at $\left(\frac{3}{2}, \frac{4}{25}\right)$



3) a)

Asymptotes: $x = -2$, $x = -1$ and $x = 2$

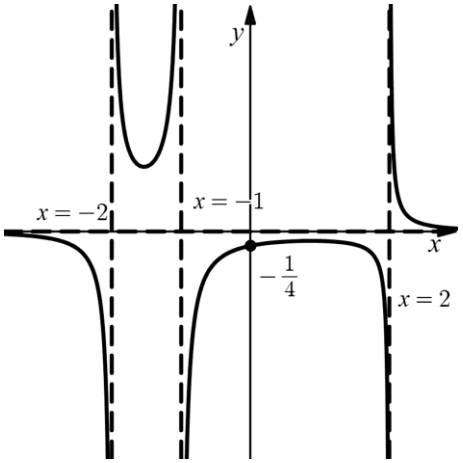
When $x = 0$, $y = -\frac{1}{4}$

As $x \to -\infty$, $\frac{1}{f(x)} \to 0^-$, as $x \to \infty$, $\frac{1}{f(x)} \to 0^+$

As $x \to -2$ from LHS $\frac{1}{f(x)} \to -\infty$, as $x \to -2$ from RHS $\frac{1}{f(x)} \to \infty$

As $x \to -1$ from LHS $\frac{1}{f(x)} \to \infty$, as $x \to -1$ from RHS $\frac{1}{f(x)} \to -\infty$

As $x \to 2$ from LHS $\frac{1}{f(x)} \to -\infty$, as $x \to 2$ from RHS $\frac{1}{f(x)} \to \infty$



3) b)

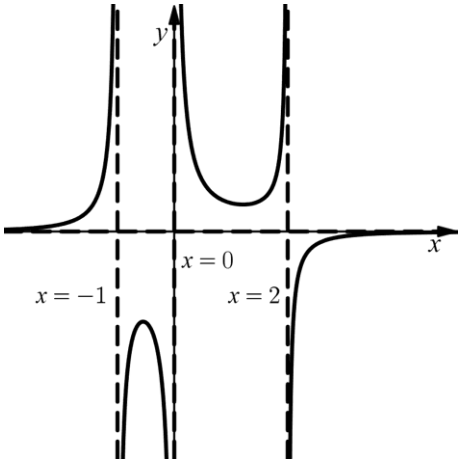
Asymptotes: $x = -1$, $x = 0$ and $x = 2$

As $x \to -\infty$, $\frac{1}{\mathrm{f}(x)} \to 0^+$, as $x \to \infty$, $\frac{1}{\mathrm{f}(x)} \to 0^-$

As $x \to -1$ from LHS $\frac{1}{\mathrm{f}(x)} \to \infty$, as $x \to -1$ from RHS $\frac{1}{\mathrm{f}(x)} \to -\infty$

As $x \to 0$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to 0$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$

As $x \to 2$ from LHS $\frac{1}{\mathrm{f}(x)} \to \infty$, as $x \to 2$ from RHS $\frac{1}{\mathrm{f}(x)} \to -\infty$



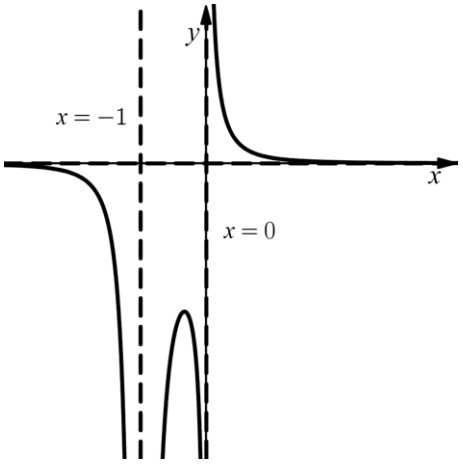
3) c)

Asymptotes: $x = -1$ and $x = 0$

As $x \to -\infty$, $\frac{1}{\mathrm{f}(x)} \to 0^-$, as $x \to \infty$, $\frac{1}{\mathrm{f}(x)} \to 0^+$

As $x \to -1$ from LHS $\frac{1}{\mathrm{f}(x)} \to \infty$, as $x \to -1$ from RHS $\frac{1}{\mathrm{f}(x)} \to -\infty$

As $x \to 0$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to 0$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$



4) a)

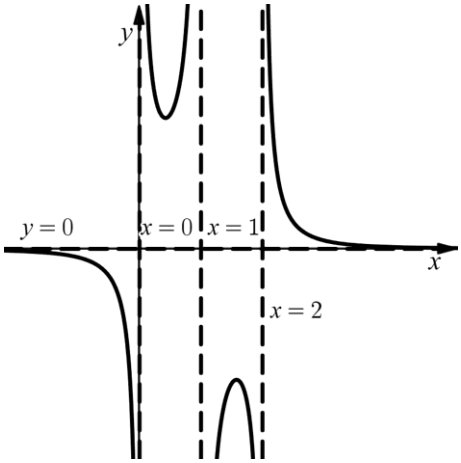
Asymptotes: $x = 0$, $x = 1$ and $x = 2$, $y = 0$

As $x \to -\infty$, $\frac{1}{\mathrm{f}(x)} \to 0^-$, as $x \to \infty$, $\frac{1}{\mathrm{f}(x)} \to 0^+$

As $x \to 0$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to 0$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$

As $x \to 1$ from LHS $\frac{1}{\mathrm{f}(x)} \to \infty$, as $x \to 1$ from RHS $\frac{1}{\mathrm{f}(x)} \to -\infty$

As $x \to 2$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to 2$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$



4) b)

$2x^3 - 9x^2 + 3x + 4 = (2x + 1)(x - 1)(x - 4)$

Asymptotes: $x = -\frac{1}{2}$, $x = 1$ and $x = 4$, $y = 0$

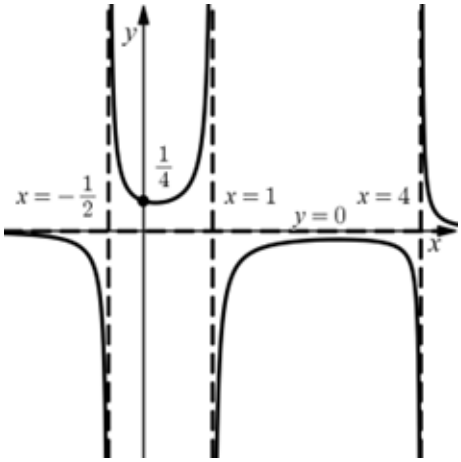
When $x = 0$, $y = \frac{1}{4}$

As $x \to -\infty$, $\frac{1}{\mathrm{f}(x)} \to 0^-$, as $x \to \infty$, $\frac{1}{\mathrm{f}(x)} \to 0^+$

As $x \to -\frac{1}{2}$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to -\frac{1}{2}$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$

As $x \to 1$ from LHS $\frac{1}{\mathrm{f}(x)} \to \infty$, as $x \to 1$ from RHS $\frac{1}{\mathrm{f}(x)} \to -\infty$

As $x \to 4$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to 4$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$



4) c)

$8 - 2x - 5x^2 - x^3 = -(x-1)(x+2)(x+4)$

Asymptotes: $x = -4$, $x = -2$ and $x = 1$, $y = 0$

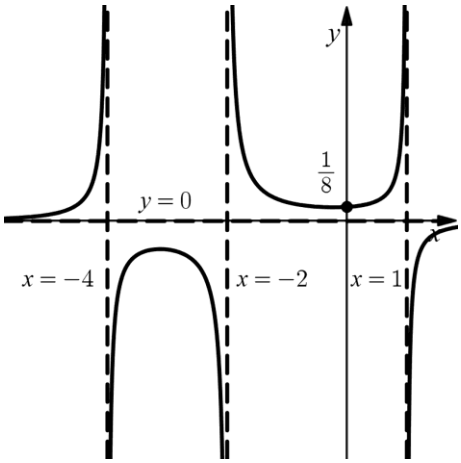
When $x = 0$, $y = \frac{1}{8}$

As $x \to -\infty$, $\frac{1}{\mathrm{f}(x)} \to 0^+$, as $x \to \infty$, $\frac{1}{\mathrm{f}(x)} \to 0^-$

As $x \to -4$ from LHS $\frac{1}{\mathrm{f}(x)} \to \infty$, as $x \to -4$ from RHS $\frac{1}{\mathrm{f}(x)} \to -\infty$

As $x \to -2$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to -2$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$

As $x \to 1$ from LHS $\frac{1}{\mathrm{f}(x)} \to \infty$, as $x \to 1$ from RHS $\frac{1}{\mathrm{f}(x)} \to -\infty$



4) d)

$x^3 + x^2 = x^2(x + 1)$

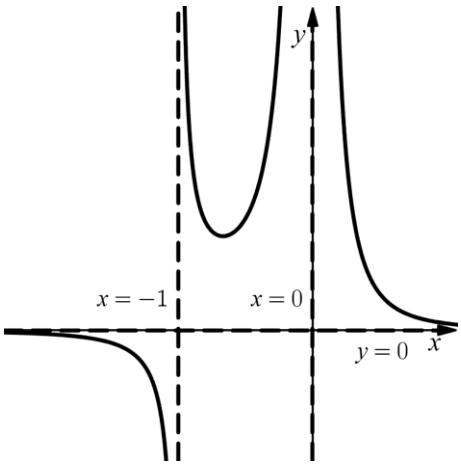
Asymptotes: $x = -1$ and $x = 0$, $y = 0$

When $x = 0$, $y = 1$

As $x \to -\infty$, $\frac{1}{\mathrm{f}(x)} \to 0^-$, as $x \to \infty$, $\frac{1}{\mathrm{f}(x)} \to 0^+$

As $x \to -1$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to -1$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$

As $x \to 0$ from LHS $\frac{1}{\mathrm{f}(x)} \to \infty$, as $x \to 0$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$



5) a)

$y = \frac{1}{k(x-1)(x-2)}$

Given $\left(\frac{3}{2}, -\frac{2}{5}\right)$ then $-\frac{2}{5} = \frac{1}{k\left(\frac{3}{2}-1\right)\left(\frac{3}{2}-2\right)} \therefore k = 10$

So $10(x - 1)(x - 2) = 10x^2 - 30x + 20$

$y = \frac{1}{10x^2-30x+20}$

$\therefore a = 10$, $b = -30$ and $c = 20$

5) b)

$y = \frac{1}{k(x+1)(x-3)}$

Given $\left(1, \frac{1}{8}\right)$ then $\frac{1}{8} = \frac{1}{k(1+1)(1-3)} \therefore k = -2$

So $-2(x + 1)(x - 3) = -2x^2 + 4x + 6$

$y = \frac{1}{-2x^2+4x+6}$

$\therefore a = -2$, $b = 4$ and $c = 6$

6)

$y = \frac{1}{k(x+1)(x-1)(x-2)}$

Given $\left(0, \frac{1}{2}\right)$ then $\frac{1}{2} = \frac{1}{k(0+1)(0-1)(0-2)} \therefore k = 1$

So $(x+1)(x-1)(x-2) = x^3 - 2x^2 - x + 2$

$y = \frac{1}{x^3-2x^2-x+2}$

$\therefore a = 1, b = -2, c = -1$ and $d = 2$

7)

$\frac{1}{\mathrm{f}(x)} = \frac{x^2-4x-5}{x^2+4x} = \frac{(x+1)(x-5)}{x(x+4)}$

Asymptotes: $y = 1, x = 0, x = -4$

$x^2 - 4x - 5 = 0 \Rightarrow x = 5, x = -1$ so intercepts: $(5, 0)$ and $(-1, 0)$

No vertical intercept.

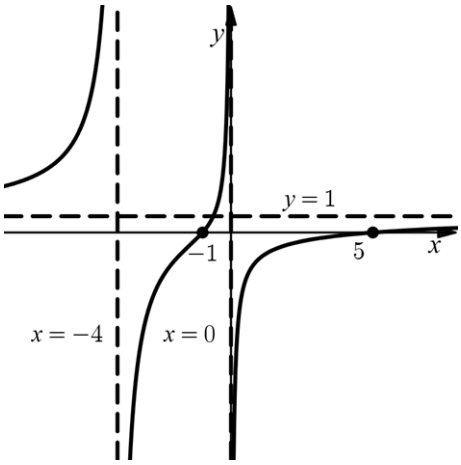
When $x \to \infty$, $y < 1$, when $x \to -\infty$, $y > 1$

At $x = -4^-$, y is large positive

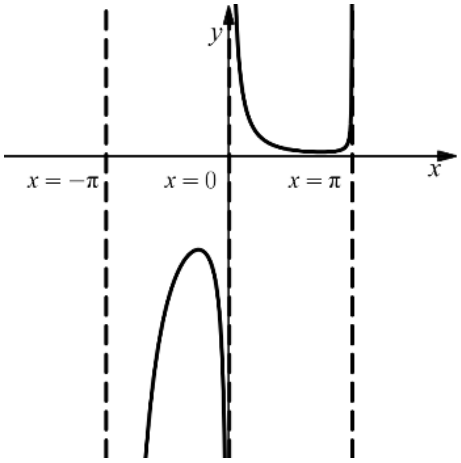
At $x = -4^+$, y is large negative

At $x = 0^-$, y is large positive

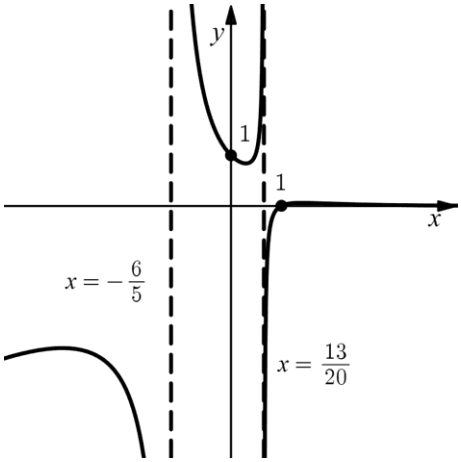
At $x = 0^+$, y is large negative



8) a)



8) b)



9)

Asymptotes at $x = 0$ and $x = \frac{1}{2}$

So $\ln(bx) = 0 \Rightarrow bx = 1 \Rightarrow x = \frac{1}{b} \therefore b = 2$

$y = \frac{1}{ax \ln 2x} \Rightarrow -\frac{e}{4} = \frac{1}{a(\frac{1}{2e}) \ln(\frac{2}{2e})} \Rightarrow -\frac{e}{4} = \frac{2e}{-a} \therefore a = 8$

$\therefore a = 8$ and $b = 2$